are three tabs in the System Assembly View, namely, the Bus Interfaces, Ports and the Addresses. The bus interface tab shows the following information for each core:

(i) Instance name (name of the core used to reference the core in the UCF)

(ii) Bus name (the bus to which the core is connected)

(iii) IP type (name of the IP core as in the IP catalogue)

(iv) IP version (the version of the IP core)

The Ports tab shows the user accessible ports of the IP cores and the nets to which they connect. The Addresses tab shows how the IP cores are mapped to the PLBs in the design

(F) is the Filters Pane area. It allows you to filter what is shown in the System Assembly View for simplified view of the design or in the case of many IP cores in the design.

16. Since we are using custom board for designing, the UCF file needs to be updated. Allot the FPGA pins to the LEDs, Clock and RST signals. The UCF file created for the ML501 board is shown in Fig. 12.

17. The download.cmd file is changed, as shown in Fig. 13, as the FPGA is the 4th device in the JTAG chain used. (The number assigned depends upon the JTAG being used.) Save the project

18. From the XPS menu, select Project → Export Hardware Design to SDK

19. In the dialogue box that appears (Fig. 14), make sure that Include Bitstream and BMM File are ticked and click Export Only

20. The bitstream will be generated and the project will be exported to SDK subsequently. Once it is done, we can progress to the next step

21. Open SDK by selecting Start → Programs → Xilinx ISE Design Suite 13.4_1 → EDK → Xilinx Software Development Kit

22. The first thing that the designer has to do is to enter the SDK workspace. SDK workspace is a folder wherein the software application(s) for a particular EDK hardware design are managed. The folder specified is shown in Fig. 15. You can choose any folder but it is better if the software folder is created inside the main project folder for ease of use

23. SDK opens with a Welcome screen. Select File → New → Xilinx C Project. Next, you have to specify the hardware platform. Click Specify on the window that appears. In the dialogue box that appears, type the name of the project as edk and use the Browse button under the Target Hardware Specification (refer Fig. 16) heading to navigate to C:\edk\SDK\SDK_Export\hw\system.xml file.

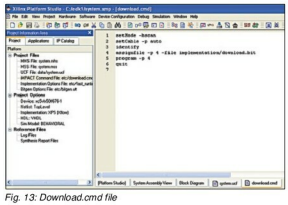


Fig. 13: Download.cmd file




Fig. 14: Export to SDK/Launch SDK




Fig. 15: Workspace Launcher

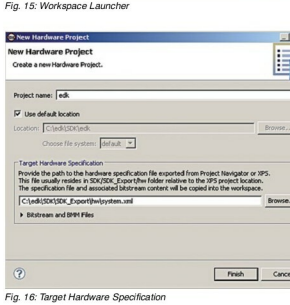



Fig. 16: Target Hardware Specification

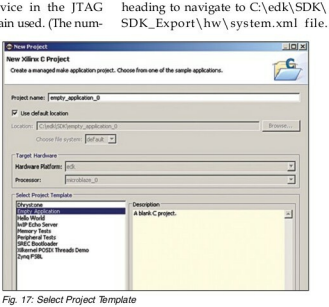



Fig. 17: Select Project Template